

- Q.28 Write a short on fuse and its uses. (CO-8)
- Q.29 List possible causes of electric shock and discuss its treatment. (CO-7)
- Q.30 Define motor? List six applications of induction motors. (CO-6)
- Q.31 Classify semiconductors according to energy band theory. (CO-9)
- Q.32 Define transistors and their types. (CO-9)
- Q.33 Draw & explain V-I characteristics of P-N junction diode. (CO-9)
- Q.34 Write a note on Autotransformer. (CO-4)
- Q.35 State and explain Faraday's Law of electromagnetic induction? (CO-1)

SECTION-D

Note: Long answer type questions. Attempt any two out of three questions. (2x10=20)

- Q.36 What are various wiring arrangements in an electrical installation? Give merits and demerits of each system. (CO-7)
- Q.37 What is the power factor and how can it be improved? (CO-3)
- Q.38 Explain the following : (CO-9)
- i) SCR ii) Servo motor

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4th Sem. / Plastic

**Subject : Basics of Electrical and
Electronics Engineering**

Time : 3 Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice Questions. All questions are compulsory (10x1 = 10)

- Q.1 Frequency of A.C. used in India is equal to (CO-1)
- a) 20Hz b) 40Hz
- c) 50Hz d) 60Hz
- Q.2 Unit of magnetomotive force is (CO-4)
- a) Weber b) Tesla
- c) Amp-turns d) Reluctance
- Q.3 The resistance of wire varies inversely as (CO-1)
- a) Area of cross-section
- b) Length
- c) Resistivity d) Temperature

Q.4 Transformer core lamination is made up of _____ (CO-4)

- a) Silicon steel b) Cast steel
c) Cast iron d) Aluminium

Q.5 The frame of an induction motor is made of _____ (CO-6)

- a) Aluminium b) Silicon steel
c) Cast iron d) Stainless steel

Q.6 The unit of electric current is: (CO-1)

- a) volt b) Ampere
c) Watt d) Joule

Q.7 Maximum value of power factor is _____ (CO-8)

- a) 0.5 b) 1
c) 2 d) 0

Q.8 Two resistance of 2 ohm and 6 ohm are connected in series. Total resistance will be. (CO-2)

- a) 2 ohm b) 6 ohm
c) 4/3 ohm d) 8 ohm

Q.9 A pentavalent impurity has _____ valence electrons. (CO-9)

- a) 1 b) 2
c) 3 d) 5

Q.10 A SCR has _____ terminals (CO-9)

- a) 2 (b) 3
(c) 4 (d) 5

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Unit of current is _____ (CO-2)

Q.12 Define insulator. (CO-1)

Q.13 Define voltage (CO-2)

Q.14 Define frequency. (CO-3)

Q.15 Explain E.M.F. _____ (CO-3)

Q.16 What is PVC? (CO-7)

Q.17 Draw symbol of DIAC. (CO-9)

Q.18 What is SCR? (CO-9)

Q.19 Give one advantage of electrical energy? (CO-2)

Q.20 Define phase difference. (CO-3)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 State and explain Kirchoff's law (CO-2)

Q.22 Give the application of D.C. (CO-1)

Q.23 Define capacitor and give its types. (CO-2)

Q.24 Explain Lenz's law. (CO-5)

Q.25 Explain high voltage and low voltage distribution systems. (CO-5)

Q.26 State the properties of a good fuse element. (CO-7)

Q.27 Define earthing and what is the purpose of earthing. (CO-8)

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